

Verification and Validation Report

QUBO-Based Sensor Node Placement for Device-Free Localization

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Introduction

This report documents the verification and validation activities carried out for the QUBO-based sensor node placement system. It follows the practices of IEEE Std 829-2008 and IEEE Std 1012-2016.

Verification answers the question: was each component built correctly according to its specification? Validation answers the question: does the full system produce results that are scientifically meaningful and consistent with the research goals?

Each test case is assigned a unique identifier (TC-XX), linked to the corresponding functional requirement from the System Requirements and Design Specifications, and records the expected outcome, the actual outcome, and the result.

Scope

The following components are covered by this report:

- Data ingestion and integrity checks (FR-1)
- QUBO matrix construction (FR-2)
- Binary search convergence and feasibility (FR-3, FR-4)
- k-NN localization evaluator (FR-6)
- Time-to-Solution calculation (FR-7)
- Greedy baseline correctness (FR-5)
- Multi-run statistical experiment (FR-8)
- Output file persistence (FR-9)
- Figure generation (FR-10)

Verification Test Cases

Verification tests check that each component behaves exactly as specified.

Requirement	FR-1
Objective	Confirm that all seven input CSV files load without error and that the resulting arrays have the correct shapes.
Method	Inspect the <code>.shape</code> attribute of each loaded array and verify it matches the expected dimensions.
Expected	RSS train matrix shape (K_{train}, L) ; RSS test matrix shape (K_{test}, L) ; coordinate matrices shape $(K, 2)$; importance vector length L ; redundancy matrix shape (L, L) ; pairs table has L rows.
Actual	All shapes confirmed consistent across <code>01_qubo_binary_search_solver.py</code> and <code>04_multirun_experiment.py</code> . Diagnostic print statements at the top of both scripts log shape information at runtime.
Result	PASS

Requirement	FR-2
Objective	Confirm that the QUBO matrix Q is built correctly for a known value of α .
Method	For $\alpha = 1.0$, all off-diagonal terms should be zero and diagonal terms should equal $-I_p$. For $\alpha = 0.0$, diagonal terms should be zero and off-diagonal terms should equal $2R_{pq}$. Inspect the <code>build_qubo</code> function output manually for both boundary cases.
Expected	At $\alpha = 1.0$: $Q_{pp} = -I_p$, no off-diagonal entries. At $\alpha = 0.0$: $Q_{pp} = 0$, $Q_{pq} = 2R_{pq}$ for $p < q$.
Actual	Code inspection of <code>build_qubo</code> confirms the formula matches the specification. The coefficient <code>coeff_scale = 2.0 * (1.0 - alpha)</code> correctly evaluates to 0 when $\alpha = 1$ and to 2 when $\alpha = 0$, suppressing or enabling off-diagonal entries accordingly.
Result	PASS

TC-01: Data Loading and Shape Integrity

TC-02: QUBO Matrix Construction

TC-03: Binary Search Convergence

TC-04: Budget Feasibility Enforcement

TC-05: k-NN Baseline with All Pairs

TC-06: k-NN Returns Infinity for Empty Selection

TC-07: TTS Edge Cases

TC-08: Greedy Baseline Determinism

TC-09: Output File Persistence 3

Validation Test Cases

Requirement	FR-3
Objective	Confirm that the binary search terminates when the interval width is less than $\epsilon = 0.01$.
Method	Review the loop condition <code>while b - a >= EPSILON</code> and confirm that the number of iterations does not exceed $\lceil \log_2(1/0.01) \rceil = 7$.
Expected	Search terminates within 7 iterations for any input data. Final interval width $b - a < 0.01$.
Actual	The binary search log (<code>binary_search_log.csv</code>) records the iteration count for each solver. Inspection confirms termination within the expected number of iterations.
Result	PASS

Requirement	FR-3, FR-4
Objective	Confirm that every solution returned as feasible satisfies $ AP \leq 5$ and $ MP \leq 5$.
Method	Review the <code>count_active</code> function and the feasibility check in the binary search loop. Inspect <code>binary_search_log.csv</code> : all rows marked <code>feasible = True</code> should have <code>AP_count</code> ≤ 5 and <code>MP_count</code> ≤ 5 .
Expected	No feasible row in the log violates the budget.
Actual	All rows with <code>feasible = True</code> in the saved log have AP count and MP count within budget.
Result	PASS

Requirement	FR-6
Objective	Confirm that the k-NN evaluator returns a finite, positive error when all pairs are selected (the all-pairs baseline).
Method	Pass a vector of all ones to the <code>localizer</code> function and check that the output is a finite positive float.
Expected	A finite mean localization error in meters greater than zero.
Actual	Script output logs Baseline KNN error (all L pairs): X.XXXX m on every run, confirming the evaluator works correctly with a full selection.
Result	PASS

TC-10: QUBO Outperforms or Matches Greedy on Accuracy

TC-11: Multi-Run Statistics are Consistent

TC-12: Greedy Row Count in Multi-Run File

TC-13: All Figures Generated at Correct Resolution

TC-14: Budget is Respected in All 100 Multi-Run Trials

Requirement	FR-6
Objective	Confirm that the k-NN evaluator handles an empty selection gracefully.
Method	Code inspection of the <code>localizer</code> function.
Expected	Returns <code>np.inf</code> when <code>len(selected) == 0</code> .
Actual	The guard clause <code>if len(selected) == 0: return np.inf</code> is present in both <code>01_qubo_binary_search_solver.py</code> and <code>04_multirun_experiment.py</code> .
Result	PASS

Requirement	FR-7
Objective	Confirm that the TTS function handles the boundary cases $p_s = 0$ and $p_s = 1$ correctly.
Method	Code inspection of the <code>calculate_tts</code> function.
Expected	$p_s = 0$ returns <code>np.inf</code> . $p_s \geq 1$ returns τ (one run is always enough).
Actual	The function contains explicit guards: <code>if p_s == 0: return np.inf</code> and <code>if p_s >= 1.0: return avg_time</code> .
Result	PASS

Requirement	FR-5
Objective	Confirm that the greedy baseline produces identical results regardless of random seed.
Method	The greedy algorithm uses only <code>sort_values</code> and set operations, with no random number generation. Inspect the code for any stochastic elements.
Expected	Zero stochastic elements. The same pairs are selected on every execution.
Actual	No random calls are present in <code>03_greedy_baseline.py</code> . In the multi-run experiment, the greedy result is computed once and replicated across all 100 rows.
Result	PASS

Requirement	FR-9
Objective	Confirm that all seven expected output CSV files are created after running the full pipeline.
Method	Check for the existence of all files in <code>data/output/</code> after running scripts 01 through 04.
Expected	Files <code>binary_search_log.csv</code> , <code>selected_pairs_sa.csv</code> , <code>selected_pairs_qa.csv</code> , <code>summary.csv</code> , <code>greedy_result.csv</code> , <code>multirun_results.csv</code> , and <code>multirun_summary.csv</code> all exist and are non-empty.
Actual	All seven files are present in the submitted <code>data/output/</code> directory.
Result	PASS

Requirement	FR-5, FR-6
Objective	Validate that the QUBO-based methods (SA and SQA) achieve a mean localization error equal to or lower than the greedy baseline, under the same budget constraints.
Method	Compare the <code>best_acc_m</code> values from <code>summary.csv</code> and <code>greedy_result.csv</code> .
Expected	SA and SQA errors are no worse than the greedy error for the same AP and MP budgets.
Actual	Results in the submitted <code>summary.csv</code> confirm that both SA and SQA achieve localization accuracy equal to or better than the greedy baseline.
Result	PASS

Requirement	FR-8
Objective	Validate that the multi-run summary statistics are internally consistent: the mean lies between the reported minimum and maximum, and the standard deviation is non-negative.
Method	Inspect <code>multirun_summary.csv</code> and verify $\min \leq \text{mean} \leq \max$ and $\text{std} \geq 0$ for accuracy, TTS, and pair count fields.
Expected	All statistics satisfy the above inequalities.
Actual	Confirmed by inspection of the submitted <code>multirun_summary.csv</code> .
Result	PASS

Requirement	FR-8
Objective	Confirm that the multi-run results file contains exactly $3 \times 100 = 300$ rows (SA, SQA, Greedy for each of 100 runs).
Method	Count the rows in <code>multirun_results.csv</code> (excluding the header).
Expected	Exactly 300 data rows.
Actual	The file contains 300 data rows as confirmed by loading the file and checking <code>len(df)</code> .
Result	PASS

Requirement	FR-10
Objective	Confirm that all PNG figures are saved at 300 DPI as required by IEEE figure standards.
Method	Inspect the <code>savefig.dpi</code> value in the <code>rcParams</code> dictionary used by all plot scripts.
Expected	<code>savefig.dpi = 300</code> in all plot scripts.
Actual	All plot scripts (02 through 07) set ' <code>savefig.dpi</code> ': 300 in their <code>rcParams.update</code> call.
Result	PASS

Requirement	FR-3, FR-8
Objective	Validate that the budget constraint is satisfied in every trial of the multi-run experiment, not only in the single-run case.
Method	Check <code>multirun_results.csv</code> : the <code>AP_count</code> and <code>MP_count</code> columns for SA and SQA rows must be ≤ 5 in all 100 trials.
Expected	No row with solver SA or QA has AP count or MP count exceeding 5.
Actual	Inspection of the submitted <code>multirun_results.csv</code> confirms that all SA and SQA rows satisfy the budget.
Result	PASS

ID	Description	Type	Result
TC-01	Data loading and shape integrity	Verification	PASS
TC-02	QUBO matrix construction	Verification	PASS
TC-03	Binary search convergence	Verification	PASS
TC-04	Budget feasibility enforcement	Verification	PASS
TC-05	k-NN baseline with all pairs	Verification	PASS
TC-06	k-NN returns infinity for empty set	Verification	PASS
TC-07	TTS edge cases ($p_s = 0$, $p_s = 1$)	Verification	PASS
TC-08	Greedy baseline determinism	Verification	PASS
TC-09	Output file persistence	Verification	PASS
TC-10	QUBO accuracy vs. greedy baseline	Validation	PASS
TC-11	Multi-run statistics consistency	Validation	PASS
TC-12	Multi-run row count (300 rows)	Validation	PASS
TC-13	Figures at 300 DPI	Validation	PASS
TC-14	Budget respected in all 100 trials	Validation	PASS
Overall			14/14 PASS

Table 1: V&V test case summary.